

Detecting AI-cloned voices in Indian languages

A real-vs-fake speech classifier for English, Hindi, Tamil and Telugu — and a test of where it breaks.



Team · [Group name], 4 members

Proposal due · 11 Oct 2026

Final demo · from 21 Dec

Voice cloning is cheap. Detectors are English-first.



Easy to make

Open models now clone a voice from a short reference clip, and run on a free GPU.



Used in every language

Fraud calls impersonating relatives or officials reach people in Hindi, Tamil and Telugu, not just English.



Weak transfer

Detectors trained on one language or generator lose accuracy on new ones.

Objective: classify a 4-second clip as real or AI-generated across English + 3 Indian languages, and measure how well it holds up on languages and generators it never saw.

What exists, and what we add

Prior work

ASVspoof 2019 LA

The standard benchmark. English only, 2019-era TTS.

In-the-Wild (Müller et al., 2022)

Showed lab-trained detectors fail on real-world English fakes.

IndicSynth (ACL 2025)

4,000 h of synthetic speech in 12 Indian languages; shows language bias.

Our contribution

- 1. Newer generators.** Fakes from Indic Parler-TTS, XTTS-v2 and MMS-TTS, which existing sets do not cover.
- 2. Generalisation tests.** Leave-one-language-out and leave-one-generator-out.
- 3. Phone-call robustness.** Evaluation after 8 kHz codec compression.
- 4. Explainable demo.** Saliency over time shows which part of a clip looks fake.

Five sources, all openly downloadable

Dataset	Role in the project	Direct link
ASVspoof 2019 LA	English real + fake; train/dev/eval (LA.zip, 7.1 GB)	datashare.ed.ac.uk/handle/10283/3336
Kathbath (AI4Bharat)	Real speech, 1,684 h, 12 languages; we use Hindi, Tamil, Telugu	huggingface.co/datasets/ai4bharat/Kathbath (accept terms)
IndicSynth	Synthetic speech, 4,000 h, 12 Indian languages	github.com/vdivyas/IndicSynth
Our generated set	Fakes from 3 TTS models on Kathbath sentences	Built in Milestone 2; released on HF Hub
In-the-Wild	English, test only (37.9 h real + fake)	huggingface.co/datasets/mueller91/In-The-Wild

Target: ~10k clips per language, balanced real/fake, speaker-disjoint train/dev/test splits, all resampled to 16 kHz mono.

Three detectors, three generators

BASELINE

AASIST-L

Graph attention on the raw waveform; ~85k params; pretrained ASVspoof weights.

github.com/clovaai/aasist

MAIN MODEL

XLS-R 300M + head

Self-supervised encoder pretrained on 128 languages, incl. Indian ones; attentive pooling + MLP.

[facebook/wav2vec2-xls-r-300m](https://facebook.com/wav2vec2-xls-r-300m)

COMPARISON

IndicWav2Vec

AI4Bharat encoder pretrained on Indian speech. Does Indic pretraining beat multilingual?

github.com/AI4Bharat/IndicSUPERB

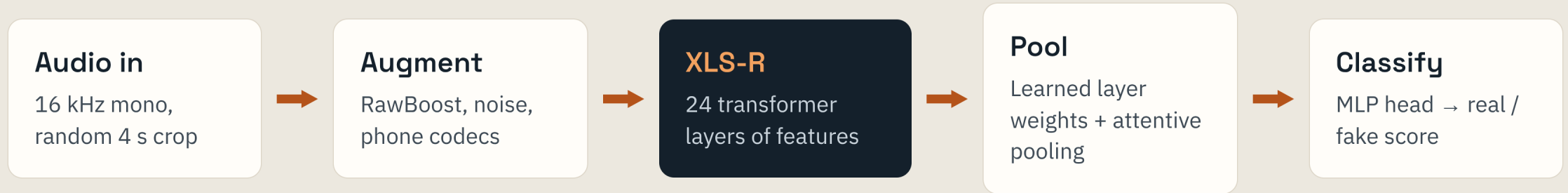
Fake generators (GenAI side)

Indic Parler-TTS · Apache-2.0 ·
ai4bharat/indic-parler-tts

XTTS-v2 (voice cloning) · non-
commercial licence · coqui/XTTS-v2

MMS-TTS · CC-BY-NC 4.0 ·
facebook/mms-tts-hin (+tam, tel)

End-to-end pipeline



Why this design

Spoof cues live in lower layers, language content in upper ones; learned layer weights let the model pick, and the weights themselves become an analysis result.

At inference

Long clips are split into 4 s windows and scores averaged; Integrated Gradients gives a per-frame saliency curve for the demo.

Unfreeze gradually, then sweep

STAGE 1

Frozen probe

Encoder frozen; train layer weights + head. LR $1e-3$, ~5 epochs.

STAGE 2

Partial fine-tune

Unfreeze top 8 layers. LR $1e-5$ encoder, $1e-4$ head.

STAGE 3

Full fine-tune

All layers, layer-wise LR decay 0.9; fp16 + gradient checkpointing.

Fixed settings

- AdamW, weight decay $1e-4$
- Cosine schedule, 10% warmup
- Batch $8 \times$ grad-accum 4
- Class-weighted cross-entropy

Sweep (small grid)

- Encoder LR: $5e-6$, $1e-5$, $3e-5$
- Unfrozen layers: 0, 8, 24
- Head dropout: 0.1, 0.3
- Augmentation on / off

Discipline

- Early stop on dev EER (patience 3)
- 3 seeds per final config
- Configs + logs in W&B / TensorBoard
- Free Kaggle / Colab GPUs (16 GB)

Five test protocols, one primary metric

Protocol	Train on	Test on	Question it answers
In-domain	All languages, all generators	Held-out speakers	Does it work at all?
Cross-language	3 of 4 languages	The 4th language	Does it transfer to a new language?
Unseen generator	2 of 3 TTS models	The 3rd model	Does it catch new attacks?
In-the-Wild	Full training set	In-the-Wild (English)	Does it survive real-world audio?
Phone channel	Full training set	Test set at 8 kHz codec	Does it survive a phone call?

Metrics: Equal Error Rate (EER, primary), ROC-AUC; min t-DCF on ASVspoof for comparison with published numbers.

Error analysis: EER broken down by language, generator, gender and clip length; saliency on the worst misses.

Six milestones to a live demo

Milestone	Due	Deliverable	Lead
M1 · Problem & lit review	22 Oct	Objectives, baselines, gap; start TTS generation	All · C
M2 · Dataset	29 Oct	Real + fake clips, speaker-disjoint splits, data card	A
M3 · Architecture	5 Nov	AASIST-L reproduced; XLS-R pipeline end-to-end	B + C
M4 · Training	19 Nov	Stages 1–3, augmentation, sweep logs	C · B
M5 · Evaluation	26 Nov	5 protocols × 3 models, error analysis	D
M6 · Deployment	10 Dec	Gradio app on HF Spaces, docs, final report	D · All
Final presentation	21 Dec+	Live demo, code walkthrough, report	All

Watch-out: M2 is only one week after M1, so fake-speech generation starts during M1 week. M4 has two weeks — the longest block, used for the sweep.

One owner per stage, everyone presents

A

Data & generation

- Download, resample, split
- Run 3 TTS generators
- Data card on HF Hub

B

Baselines

- Reproduce AASIST-L
- IndicWav2Vec variant
- Shared data loader

C

Main model

- XLS-R + head
- 3-stage fine-tuning
- Hyperparameter sweep

D

Eval & deploy

- 5 test protocols
- Saliency + error analysis
- Gradio app on HF Spaces

Shared: one GitHub repo (training scripts, inference notebooks, docs) with a weekly **WORKLOG.md** of individual contributions; fixed seeds and configs so every number is reproducible.